Week 4 Assignment: Visualization Analysis

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Comparing Visualizations of State Higher Education Funding

Our analytics team built three visualizations from the same NCES Table 333.30 dataset, which tracks state and local appropriations per full-time equivalent (FTE) student for public degree-granting postsecondary institutions from 1990-91 through 2020-21. Because the report's audience consists of policymakers and education advocates without formal data training, the goal of this evaluation is to identify the visualization that communicates the data's key insights with the least cognitive effort.

How Each Visualization Represents the Data

The first visualization ("simple") represents a single-year snapshot, displaying appropriations per FTE student for ten states in academic year 2020-21 using vertical blue bars, with a red dashed line marking the national average of $8,750. The second visualization ("technical") represents two related perspectives stacked vertically: a horizontal bar chart ranking ten states by the percent change in appropriations between 1990-91 and 2020-21, and a box plot displaying the distribution of state appropriations across six selected academic years. The third visualization ("story") tracks five large states—California, New York, Texas, Florida, and Illinois—as color-coded lines plotted across six academic years on a single time-series axis.

Key Characteristics

The simple visualization is a vertical bar chart paired with a horizontal reference line, arranged as a single panel with ten bars on a clean white background. Its labeling includes the title "State Higher Education Funding 2020-21," an x-axis of rotated state names, a y-axis labeled "Appropriations per FTE Student ($)" in $2,000 increments, and a legend identifying the national average line. The technical visualization is a multi-chart figure combining a horizontal bar chart and a box plot in two stacked panels. The bar chart is sorted from greatest to least percent change, and the box plot is arrayed left to right across six academic years. Labeling includes individual panel titles, state names beside each bar, an x-axis label of "Percent Change," and a y-axis label of "Appropriations per FTE Student ($)," though it omits annotations explaining what the boxes, whiskers, and median lines represent. The story visualization is a multi-series line chart with markers at each measured year, using a single wide panel with dashed gridlines and five color-coded lines. Labeling includes the title "30-Year Trend of State Appropriations for Higher Education by Selected States," an x-axis of six rotated year labels, a y-axis labeled "Appropriations per FTE Student ($)," and a legend mapping each color to a state.

Comparing Effectiveness: Clarity and Data Type Alignment

In terms of clarity, the simple visualization is the most immediately readable because it presents a single variable across a single time period. The story visualization is also highly readable; its color-coding, gridlines, and point markers allow a viewer to trace each state's trajectory without prior training. The technical visualization is the least accessible because its box plot panel presumes familiarity with quartiles, medians, and whiskers—concepts the target audience is unlikely to share. When it comes to how the data actually fits the visuals, we have to look at the "bones" of the information—which is a long-term record of dollar values across different states and years. The Simple Visualization falls short because it ignores the big picture, essentially taking a years-long story and reducing it to a single snapshot. The Technical Visualization is a bit more sophisticated; it uses box plots and percentages to show shifts, but it still favors statistics over the actual timeline. The Story Visualization is the clear winner here. By using a line chart, it respects the natural flow of the data, keeping the focus on how things changed over time without losing the specific details of each state.

Recommendation for the Audience

The story visualization is the most appropriate choice for the report's audience. Policy questions about funding are almost always trend questions: stakeholders want to know whether support has grown, eroded, or stagnated, and over what time horizon. The line chart answers that question directly and reveals patterns invisible in the other two charts, including the steep decline between 2000-01 and 2010-11 that coincides with the 2008 recession. The simple visualization, by contrast, strips away the historical context policymakers need, and the technical visualization risks alienating non-expert readers.

Recommended Refinements

Several refinements would strengthen the visualizations. First, the story visualization should label each line directly at its right endpoint in addition to the legend, reducing back-and-forth eye movement. Second, its y-axis values should be formatted with dollar signs and thousands separators (e.g., $7,500) to reinforce the unit at a glance. Third, an annotation near the 2010-11 trough—such as "Post-2008 recession funding decline"—would help readers connect the visual pattern to a familiar policy event. Fourth, the simple visualization should sort its bars from highest to lowest funding and display each dollar value on top, removing the need for axis estimation. Fifth, the technical visualization’s box plot should be replaced with a shaded min-to-max range band, which conveys the same insight without statistical vocabulary. Together, these refinements would make the chosen visualization speak more clearly to its non-expert audience.